

AZURE LOAD BALANCER PROJECT

Prepared by

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➤ Project Title

Implementation of Azure Public and Private Load Balancers with Web and Application VMs.

➤ Objective

The objective of this project is to design and implement a two-tier architecture in Azure using **Public Load Balancer** and **Private Load Balancer**. The setup ensures:

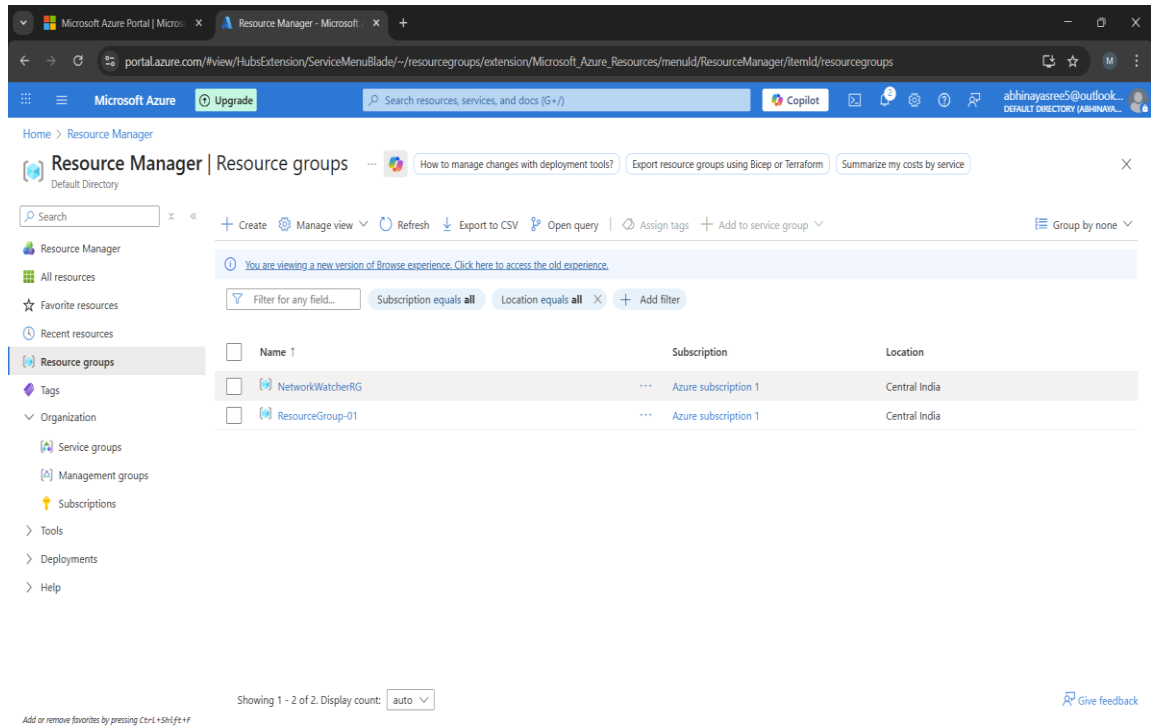
- Public access only to Web VMs through a Public Load Balancer.
- Secure internal communication between Web VMs and Application VMs through a Private Load Balancer.
- Verification of connectivity using telnet.

➤ Architecture Overview

- **Public Load Balancer**
 - Frontend: Public IP
 - Backend Pool: Web VM(s)
 - Used to expose NGINX service to the internet
- **Private Load Balancer**
 - Frontend: Private IP
 - Backend Pool: Application VM(s)
 - Used for internal communication between Web and App layers
- **Virtual Machines**
 - Web VM: Hosts NGINX
 - App VM: Hosts Apache Tomcat

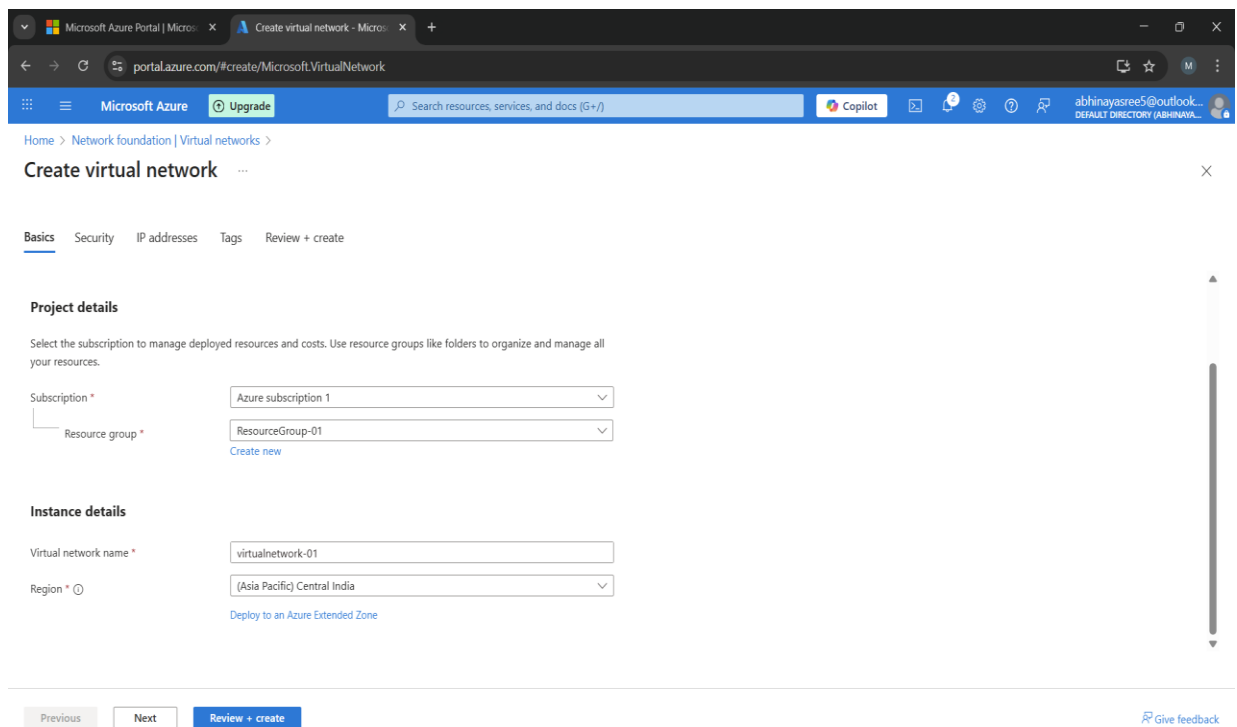
➤ Step 1: Create Virtual Network and Subnets

- Before creating Vnet create Resource group



➤ Create a Virtual Network (VNet)

- Address space: 10.0.0.0/16



- Create two subnets:
- **Web Subnet:** 10.0.0.0/24
 - **AppSubnet:** 10.0.1.0/24

The screenshot shows the 'Create virtual network' page in the Microsoft Azure Portal, specifically the 'IP addresses' tab. The page displays the '10.0.0.0/16' address space with a table of subnets. The subnets are 'abhinayasreeWEBsubnet' and 'abhinayasreeAPPsubnet', both with IP ranges 10.0.0.0 - 10.0.0.255 and 10.0.1.0 - 10.0.1.255 respectively, each having a size of /24 (256 addresses). The page includes navigation tabs for Basics, Security, IP addresses, Tags, and Review + create. A 'Review + create' button is visible at the bottom.

Subnets	IP address range	Size	NAT gateway
abhinayasreeWEBsubnet	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-
abhinayasreeAPPsubnet	10.0.1.0 - 10.0.1.255	/24 (256 addresses)	-

Create virtual network with two subnets

The screenshot shows the 'virtualnetwork-01' page in the Microsoft Azure Portal, specifically the 'Subnets' tab. The page displays a table of subnets. The subnets are 'abhinayasreeWEBsubnet' and 'abhinayasreeAPPsubnet', both with IP ranges 10.0.0.0/24 and 10.0.1.0/24 respectively, each having 251 available IPs. The page includes a search bar and a table with columns for Name, IPv4, IPv6, Available IPs, Delegated to, Security group, and Route table. The page also shows a 'Showing 2 subnets' message at the bottom.

Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
abhinayasreeWEBsubnet	10.0.0.0/24	-	251	-	-	-
abhinayasreeAPPsubnet	10.0.1.0/24	-	251	-	-	-

➤ Step 2: Create Web Virtual Machine (Web VM)

- OS: Linux (Ubuntu)
- Subnet: Web Subnet
- Public IP: Yes (for SSH access)

Microsoft Azure Portal | Create a virtual machine - Microsoft Azure

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high availability | Help me create a low cost VM

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Your resources:

Subscription *

Resource group *
[Create new](#)

Instance details

Virtual machine name *

Region *
[Deploy to an Azure Extended Zone](#)

Availability options

Security type

Image *
[See all images](#) | [Configure VM generation](#)

☒ This image is compatible with additional security features. [Click here to swap to the Trusted launch security type.](#)

VM architecture ☐ Arm64 ☒ x64

< Previous | Next : Disks > | **Review + create**

[Give feedback](#)

- Create virtual machine and allow public inbound rules HTTP (80), SSH (22)

Microsoft Azure Portal | Create a virtual machine - Microsoft Azure

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high availability | Help me create a low cost VM

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Authentication type ☐ SSH public key ☒ Password

Username *

Password *

Confirm password *

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ☐ None ☒ Allow selected ports

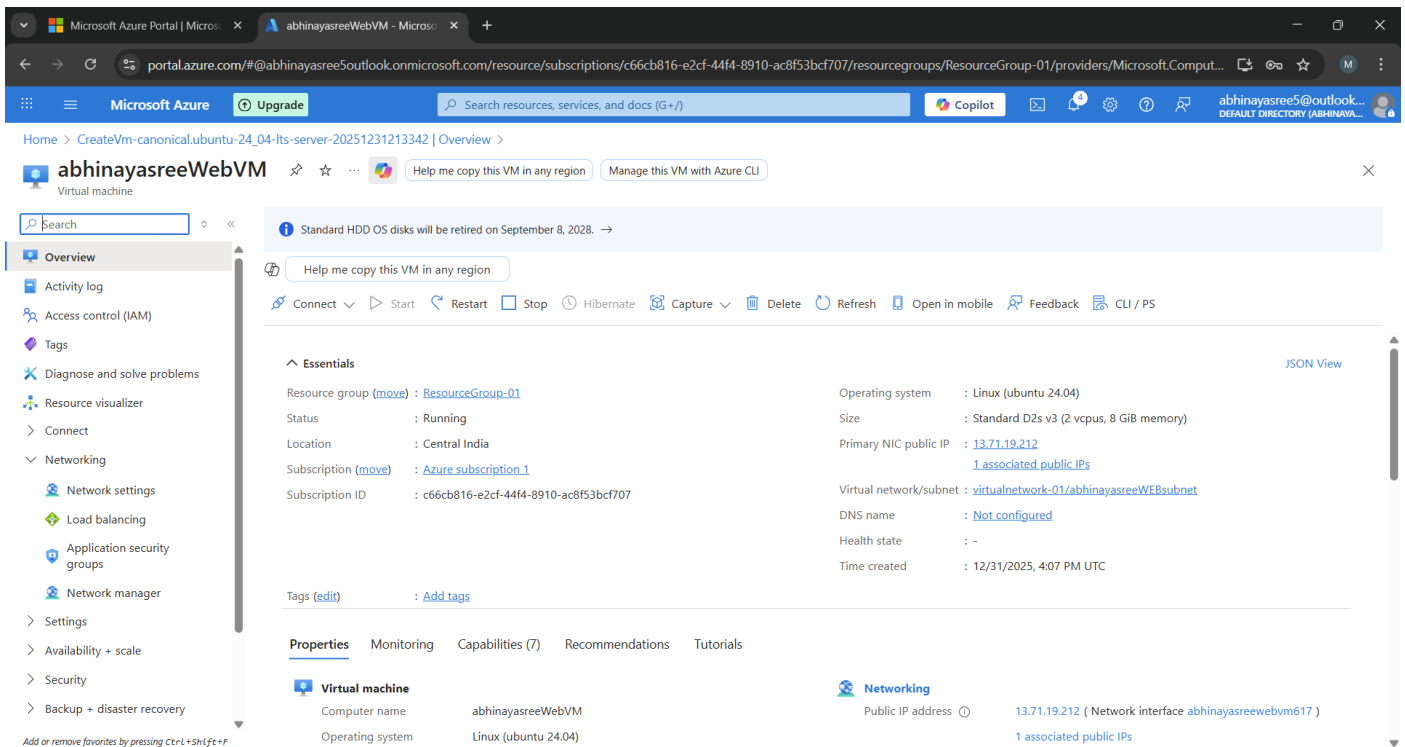
Select inbound ports *

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous | Next : Disks > | **Review + create**

[Give feedback](#)

- And click on review + create

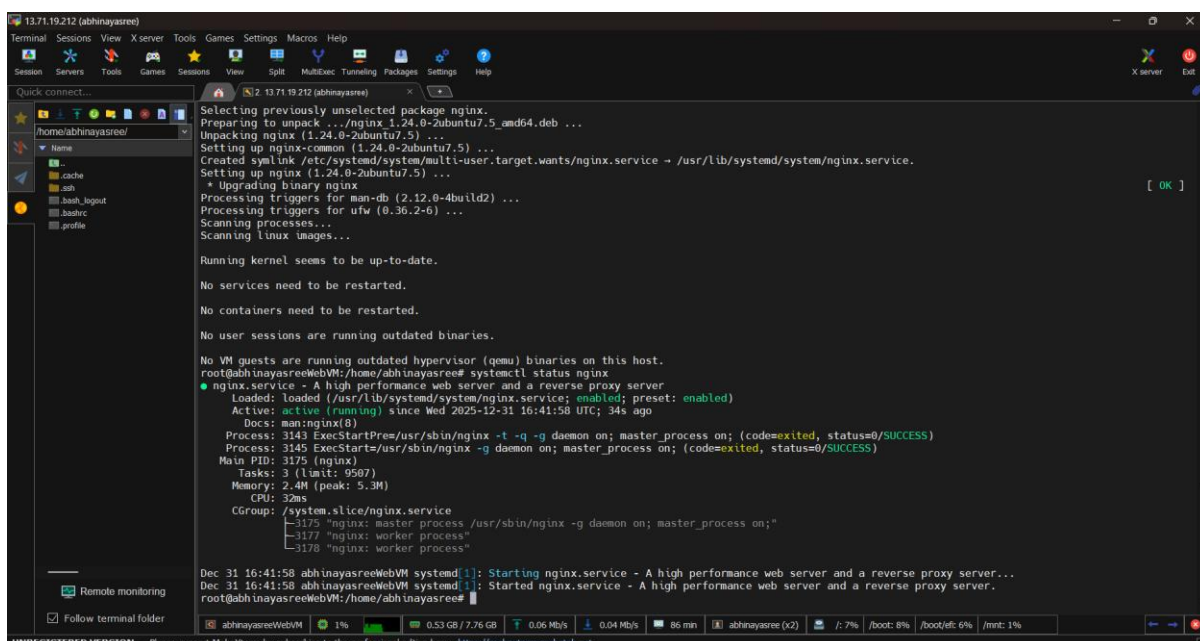


- Here the webVM is created in virtual network in web subnet

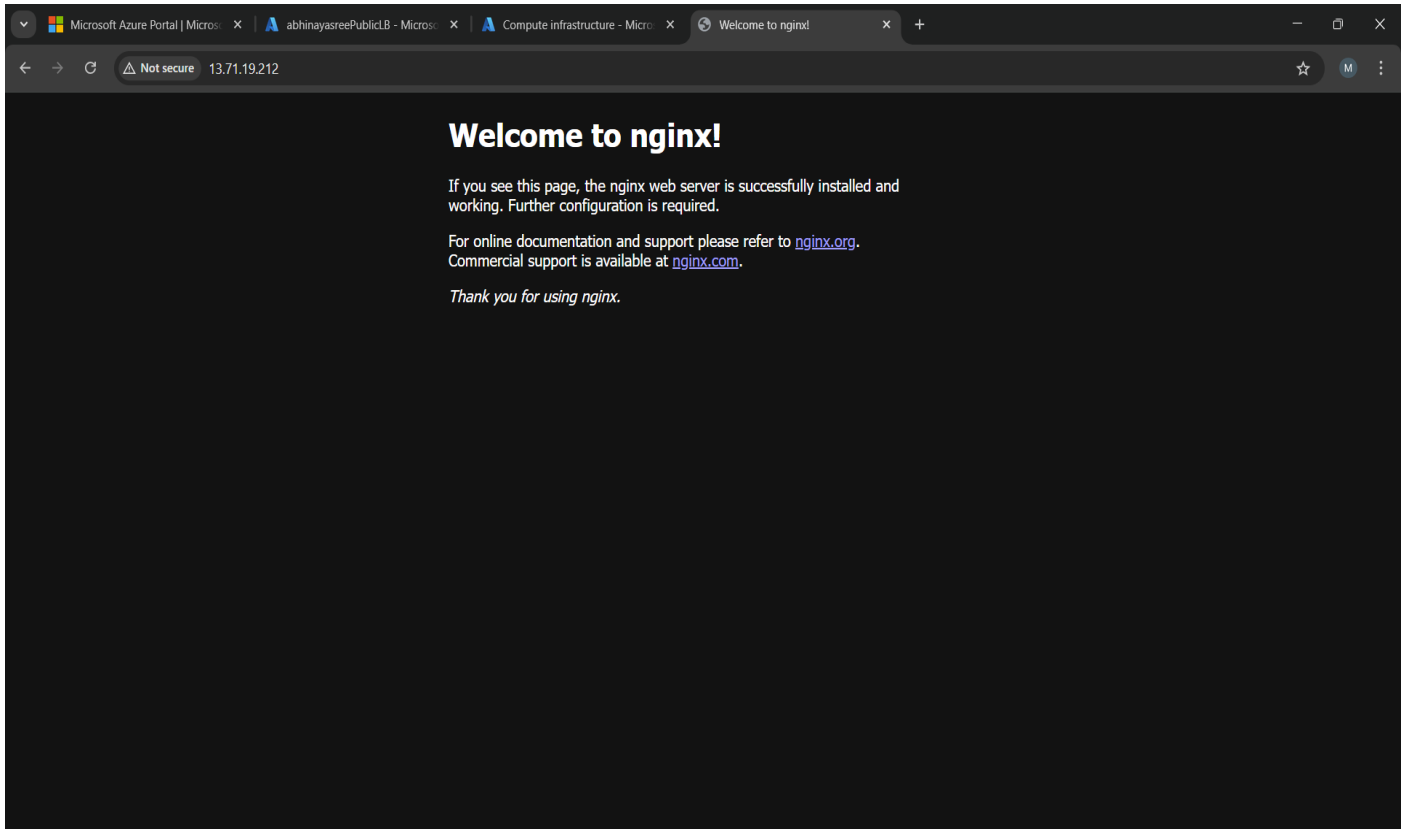
➤ Step 3: Install NGINX on Web VM

By using below commands install nginx in mobaxterm

- Sudo su (to switch to root user)
- sudo apt update
- sudo apt install nginx -y
- sudo systemctl start nginx



- The successful installation of NGINX was verified by checking the service status using the systemctl command, which showed the service as active and running in the above image.
- Verify NGINX:
- Browse Web VM public IP
 - NGINX default page should be displayed



➤ **Step 4: Create Application Virtual Machine (App VM)**

- OS: Linux (Ubuntu)
- Subnet: App Subnet
- Public IP: Optional (recommended only for admin access)

Microsoft Azure Portal | Microsoft | Create a virtual machine - Microsoft

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

Microsoft Azure Upgrade Search resources, services, and docs (G+)

abhinayasree5@outlook... DEFAULT DIRECTORY (ABHINAV...)

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me create a low cost VM Help me choose the right VM size for my workload Help me create a VM optimized for high availability

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * ResourceGroup-01 [Create new](#)

Instance details

Virtual machine name * abhinayasreeAppVM ✓

Region * (Asia Pacific) Central India [Deploy to an Azure Extended Zone](#)

Availability options No infrastructure redundancy required

Security type Standard

Image * Ubuntu Server 24.04 LTS - x64 Gen2 (free services eligible) [See all images](#) [Configure VM generation](#)

This image is compatible with additional security features. [Click here to swap to the](#)

< Previous Next : Disks > Review + create Give feedback

- Create app virtual machine and allow inbound rules SSH (22)

Microsoft Azure Portal | Microsoft | Create a virtual machine - Microsoft

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

Microsoft Azure Upgrade Search resources, services, and docs (G+)

abhinayasree5@outlook... DEFAULT DIRECTORY (ABHINAV...)

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Help me create a low cost VM Help me choose the right VM size for my workload Help me create a VM optimized for high availability

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

When creating a virtual machine, a network interface will be created for you.

Virtual network * virtualnetwork-01 [Create new](#)

Subnet * abhinayasreeAPPsubnet (10.0.1.0/24) [Manage subnet configuration](#)

Public IP * (new) abhinayasreeAppVM-ip [Create new](#)

Public IP addresses have a nominal charge. [Estimate price](#)

NIC network security group *
☐ None
☒ Basic
☐ Advanced

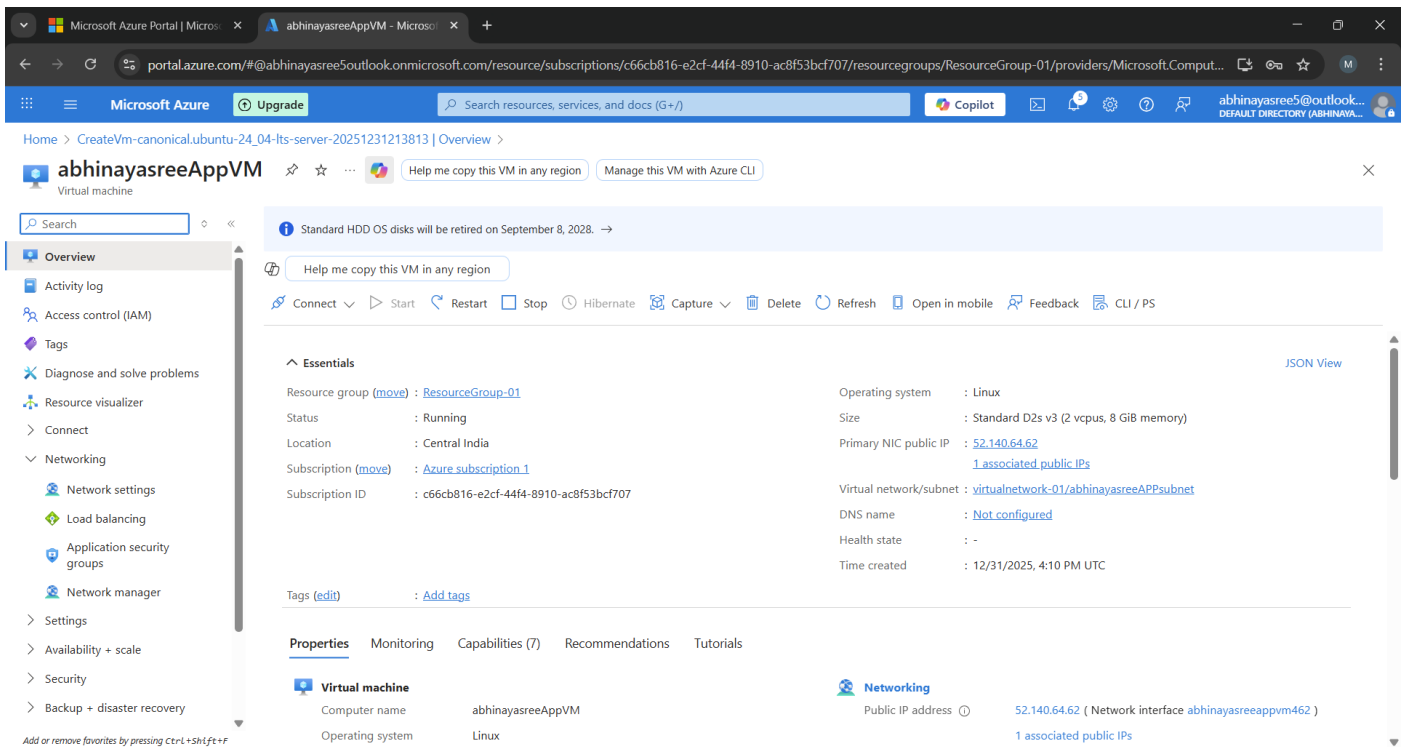
Public inbound ports *
☐ None
☒ Allow selected ports

Select inbound ports * HTTP (80), SSH (22)

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous Next : Management > Review + create Give feedback

- Click on review + create



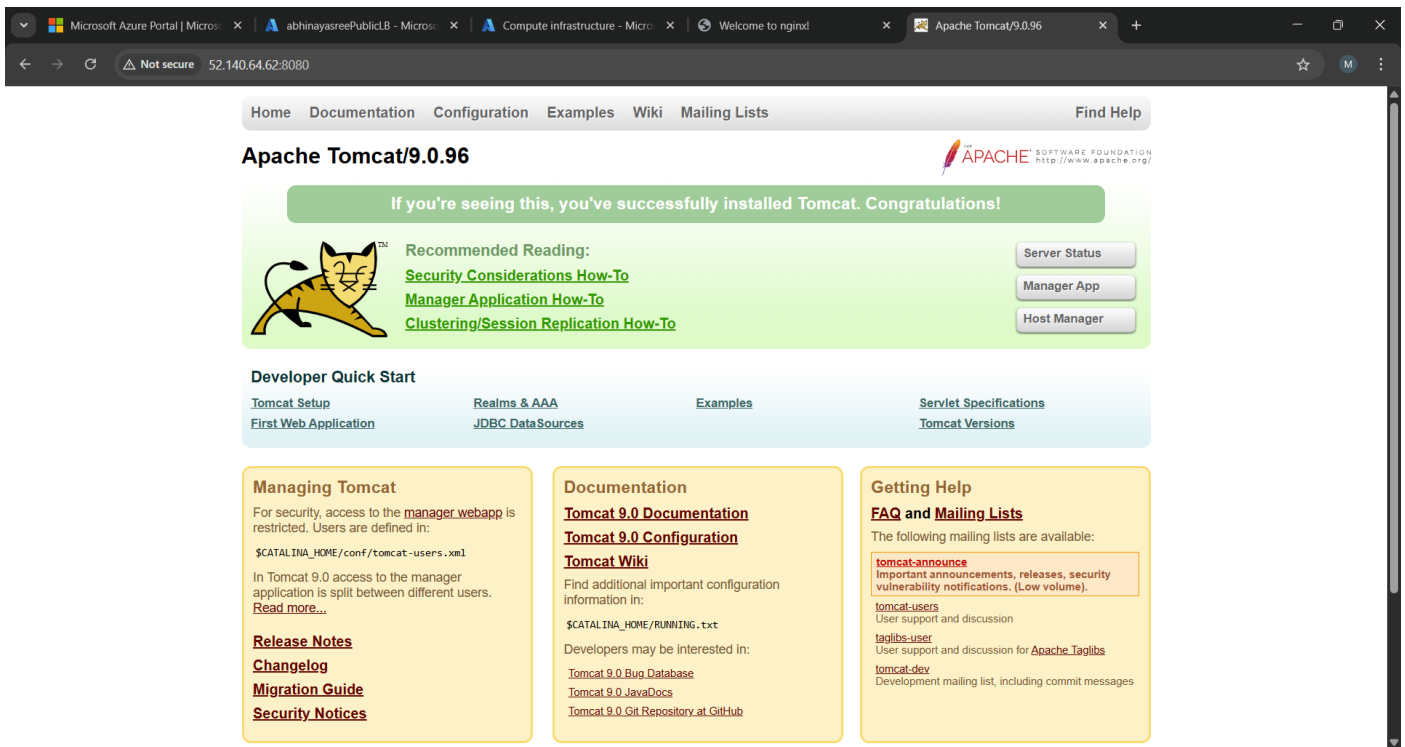
- Here the AppVM is created in virtual network in App subnet

➤ Step 5: Install Tomcat on App VM

By using below commands install Tomcat in mobaxterm

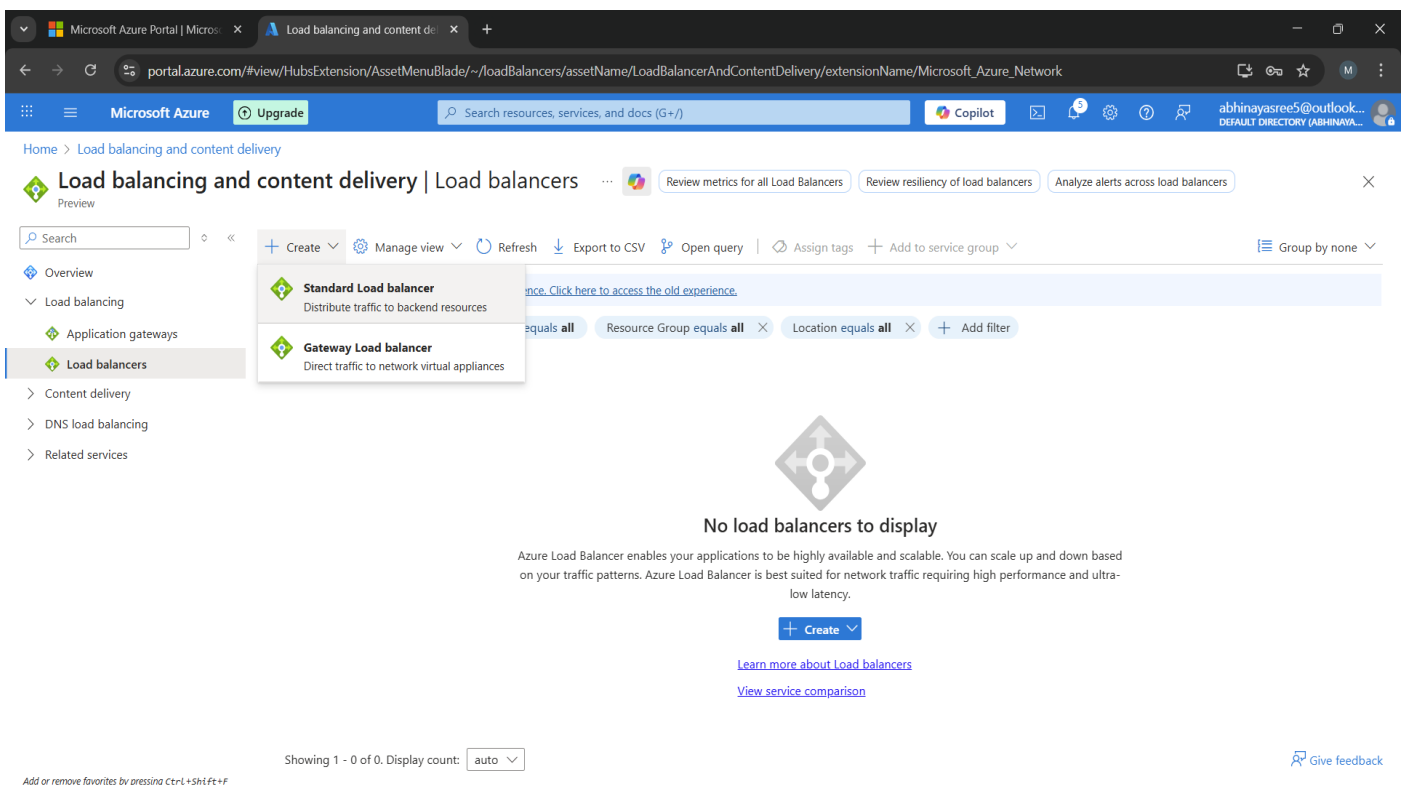
- Sudo su (to switch to root user)
- sudo apt update
- sudo apt install openjdk-11-jdk -y
- wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.96/bin/apache-tomcat-9.0.96.tar.gz
- tar -xvzf apache-tomcat-9.0.96.tar.gz
- cd apache-tomcat-9.0.96
- ls
- cd bin
- ./startup.sh

- Apache Tomcat was successfully installed and started, and the default Tomcat welcome page was verified by accessing it via the VM public IP on port 8080.



➤ Step 5: Create Azure Public Load Balancer

1. Create **Public Load Balancer**
2. Select standard load balancer



Microsoft Azure Portal | Microsoft | Create load balancer - Microsoft

portal.azure.com/#create/Microsoft.LoadBalancer-ARM

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Load balancing and content delivery | Load balancers >

Create load balancer

Standard public load balancers are accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. [Learn more.](#)

Project details

Subscription * Azure subscription 1

Resource group * ResourceGroup-01 [Create new](#)

Instance details

Name * abhinayasreepublicIP

Region * Central India

SKU ⓘ

☒ Standard (Distribute traffic to backend resources)

☐ Gateway (Direct traffic to network virtual appliances)

Type * ⓘ

☒ Public

☐ Internal

Tier *

☒ Regional

☐ Global

[Review + create](#) [< Previous](#) [Next > Frontend IP configuration](#) [Download a template for automation](#) [Give feedback](#)

- Here the type should be **Public** because it is public load balancer
- Frontend IP Configuration:

- Public IP address

3. Backend Pool:

- Add Web VM

Microsoft Azure Portal | Microsoft | Add IP configurations to backe

portal.azure.com/#view/Microsoft_Azure_Network/RegionalLBBackendPoolManage.ReactView/loadBalancerId~/null/expandedLoadBalancer~/7B"name%3A"abhinayasreePublicLB"%2C"id"...

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Load balancing and content delivery | Load balancers >

Add backend pool

Name * abhiWebPool

Virtual network ⓘ virtualnetwork-01 (Resource group: ResourceGroup-01) [The dropdown only](#)

Backend Pool Configuration

☒ NIC

☐ IP address

IP configurations

IP configurations associated to virtual machines and virtual machine scale sets must be in same location and be in the same virtual network.

[+ Add](#) | [X Remove](#)

Resource Name	Resource group	Type	IP configuration	IP Address	Availability set	Tags
Virtual machine (2)						
☐ abhinayasreeAppVM	ResourceGroup-01	Virtual machine	ipconfig1	10.0.1.4	-	-
☒ abhinayasreeWebVM	ResourceGroup-01	Virtual machine	ipconfig1	10.0.0.4	-	-

[Save](#) [Cancel](#) [Give feedback](#)

[Add](#) [Cancel](#) [Give feedback](#)

4. Health Probe:

- Protocol: TCP
- Port: 80

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, and Monitoring. The 'Health probes' option is selected. The main content area displays the 'abhinayasreePublicLB | Health probes' page. It includes a search bar, 'Add', 'Refresh', and 'Give feedback' buttons. A message states: 'To check the health status of your instances, navigate to the Load Balancing Rules page'. Below this is a table with the following data:

Name	Protocol	Port	Path	Used By
checkwebhealth	Tcp	80	-	-

5. Load Balancing Rule:

- Frontend Port: 80
- Backend Port: 80
- Backend Pool: Web VM

The screenshot shows the 'Add load balancing rule' configuration page in the Microsoft Azure portal. The page includes the following fields and values:

- Name:** loadbalancer1
- IP version:** ☒ IPv4, ☐ IPv6
- Frontend IP address:** publicfrontend (98.70.216.238)
- Backend pool:** abhiWebPool
- Protocol:** ☒ TCP, ☐ UDP
- Port:** 80
- Backend port:** 80
- Health probe:** checkwebhealth (TCP:80) (with a 'Create new' link)
- Session persistence:** None

At the bottom, there are 'Save' and 'Cancel' buttons, and a 'Give feedback' link.

- After saving you can see public load balancer

The screenshot shows the Azure Portal interface for a public load balancer. The left sidebar contains navigation options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Frontend IP configuration, Backend pools, Health probes, Load balancing rules, Inbound NAT rules, Outbound rules, Properties, Locks, and Monitoring. The main content area displays the 'Essentials' section with details such as Resource group (ResourceGroup-01), Location (Central India), Subscription (Azure subscription 1), Subscription ID (c66cb816-e2cf-44f4-8910-ac8f53bcf707), SKU (Standard), Tier (Regional), and Tags (Add tags). It also shows the Backend pool (abhiWebPool (1 virtual machine)), Load balancing rule (loadbalancer1 (Tcp/80)), Health probe (checkwebhealth (Tcp/80)), Inbound NAT rules (-), Outbound rules (-), and Frontend IP address (98.70.216.238 (publicfrontendIP)). Below this, there are three cards: 'Balance IPv4 and IPv6 addresses', 'Build highly reliable applications', and 'Secure your networks', each with a 'View' button.

➤ Step 6: Create Azure Private Load Balancer

1. Create Private Load Balancer
2. Frontend IP Configuration:

The screenshot shows the 'Add frontend IP configuration' dialog in the Azure Portal. The dialog is titled 'Add frontend IP configuration' and is for a resource named 'abhinayasreePrivateLB'. It contains the following fields:

- Name ***: PrivateFrontend
- IP version**: ☒ IPv4, ☐ IPv6
- Virtual network ***: virtualnetwork-01
- Subnet ***: abhinayasreeAPPsubnet (10.0.1.0/24)
- Assignment**: ☒ Dynamic, ☐ Static
- Availability zone ***: Zone-redundant

 At the bottom, there are 'Save' and 'Cancel' buttons, and a 'Give feedback' link. The background shows the 'Create load balancer' wizard with tabs for Basics, Frontend IP configuration, Backend pools, Inbound rules, Outbound rules, Tags, and Review + create. The 'Frontend IP configuration' tab is currently active.

- Private IP from AppSubnet

4. Health Probe:

- Protocol: TCP
- Port: 8080

The screenshot shows the Microsoft Azure portal interface. The breadcrumb navigation at the top indicates the path: Home > CreateLoadBalancerBladeV2-20251231215716 | Overview > abhinayasreePrivateLB. The main heading is 'abhinayasreePrivateLB | Health probes'. On the left, a sidebar menu lists various options, with 'Health probes' currently selected. The main content area features a search bar, '+ Add', 'Refresh', and 'Give feedback' buttons. Below these, a message states: 'To check the health status of your instances, navigate to the Load Balancing Rules page'. A table displays the health probe configuration:

Name	Protocol	Port	Path	Used By
checkapphealth	Tcp	8080	-	-

5. Load Balancing Rule:

- Frontend Port: 8080
- Backend Port: 8080
- Backend Pool: App VM

Microsoft Azure Portal | Micros... x abhinayasreePublicLB - Micros... x Add load balancing rule - Micros... x +

portal.azure.com/#view/Microsoft_Azure_Network/LoadBalancingRuleManage.ReactView/loadBalancerId/%2Fsubscriptions%2Fc66cb816-e2cf-44f4-8910-ac8f53bc707%2FresourceGroups%2F... M

Home > CreateLoadBalancerBladeV2-20251231215716 | Overview > abhinayasreePrivateLB | Load balancing rules >

Add load balancing rule

abhinayasreePrivateLB

Name * loadbalancer2

IP version * ☒ IPv4 ☐ IPv6

Frontend IP address *

Backend pool *

High availability ports ☐

Protocol ☒ TCP ☐ UDP

Port *

Backend port *

Health probe *

[Give feedback](#)

- After saving you can see private load balancer

Microsoft Azure Portal | Micros... x abhinayasreePublicLB - Micros... x abhinayasreePrivateLB - Micros... x Welcome to nginx! x Apache Tomcat/9.0.96 x +

portal.azure.com/#@abhinayasree5outlook.onmicrosoft.com/resource/subscriptions/c66cb816-e2cf-44f4-8910-ac8f53bc707/resourceGroups/ResourceGroup-01/providers/Microsoft.Network... M

Home > Load balancing and content delivery | Load balancers > abhinayasreePrivateLB

Load balancing and content delivery ...

Preview

Search

+ Create Manage view

Overview

Load balancing

Application gateways

Load balancers

Content delivery

DNS load balancing

Related services

You are viewing a new version of Browse experience. Click here to access the old experience.

Name ↑

abhinayasreePrivateLB

abhinayasreePublicLB

Showing 1 - 2 of 2. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

abhinayasreePrivateLB | Frontend IP configuration

Load balancer

Search

+ Add Refresh

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Frontend IP configuration

Backend pools

Health probes

Load balancing rules

Inbound NAT rules

Properties

Locks

Monitoring

Automation

The frontend IP address configuration of a load balancer serves as the entry point for incoming traffic to the load balancer, and the load balancer then distributes the traffic to the backend pool of virtual machines or services. [Learn more](#)

Type to start filtering...

Showing all 1 items

Name	IP address	Rules count
PrivateFrontend	10.0.1.5	1

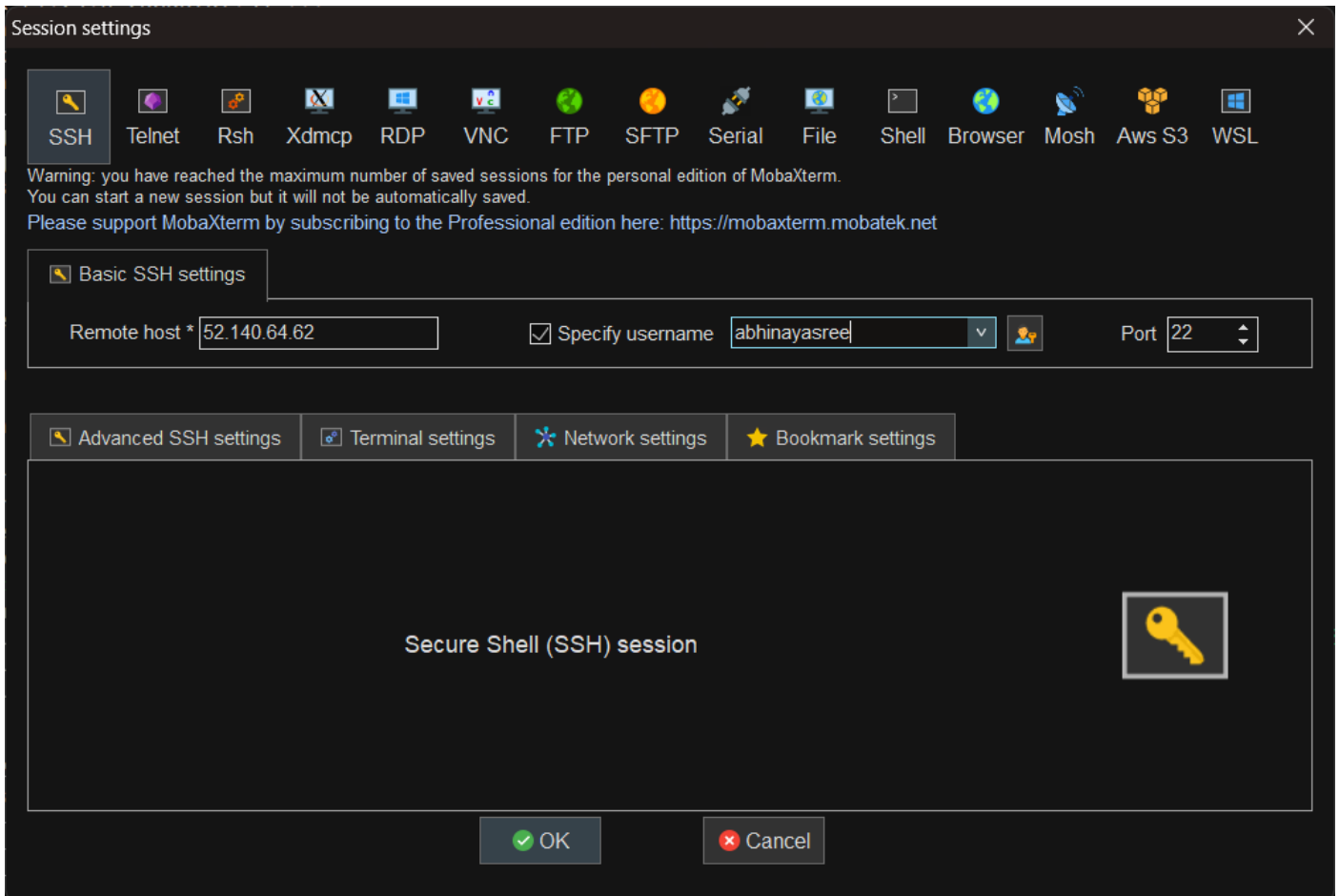
[Give feedback](#)

Add or remove favorites by pressing Ctrl+Shift+F

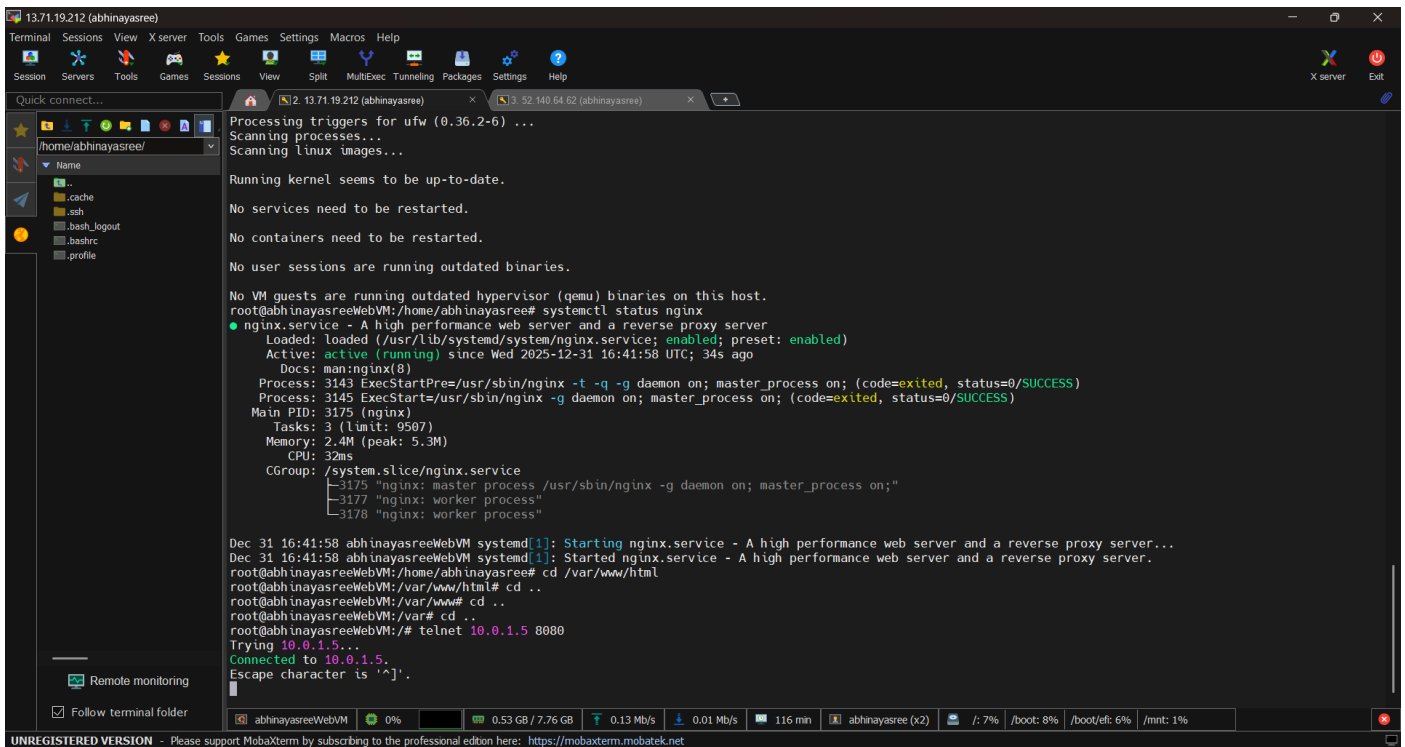
➤ Step 7: Connectivity Testing (Main Validation Step)

Test from Web VM to Private Load Balancer

- Connect to Web VM using SSH



- Give webvm public Ip and username.
- Run telnet command.



```
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@abhinayasreeWebVM:/home/abhinayasree# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-12-31 16:41:58 UTC; 34s ago
     Docs: man:nginx(8)
   Process: 3143 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 3145 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 3175 (nginx)
    Tasks: 3 (Limit: 9587)
   Memory: 2.4M (peak: 5.3M)
      CPU: 32ms
   CGroup: /system.slice/nginx.service
           └─3175 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─3177 "nginx: worker process"
               └─3178 "nginx: worker process"

Dec 31 16:41:58 abhinayasreeWebVM systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Dec 31 16:41:58 abhinayasreeWebVM systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
root@abhinayasreeWebVM:/home/abhinayasree# cd /var/www/html
root@abhinayasreeWebVM:/var/www/html# cd ..
root@abhinayasreeWebVM:/var/www# cd ..
root@abhinayasreeWebVM:/var# cd ..
root@abhinayasreeWebVM:/# telnet 10.0.1.5 8080
Trying 10.0.1.5...
Connected to 10.0.1.5.
Escape character is '^['.
```

This confirms:

- Web VM can reach App VM through Private Load Balancer.
- Internal traffic flow is working correctly.

➤ Security Validation

- Tomcat is not exposed publicly.
- Only Private Load Balancer IP is accessible from Web VM.
- Application layer is protected inside the VNet.
- End users cannot directly access Tomcat from the internet.
- This prevents unauthorized external access to the application layer.
- External traffic cannot reach the Private Load Balancer.
- Only the Web VM (NGINX) is exposed through the Public Load Balancer.
- Backend Application services remain isolated from the internet.

➤ **Conclusion:**

In this project, we successfully implemented Azure Public and Private Load Balancers to design a secure two-tier architecture. The Public Load Balancer exposes the Web layer (NGINX) to the internet, while the Private Load Balancer ensures secure internal communication with the Application layer (Tomcat). Connectivity was validated using telnet, confirming proper backend pool configuration and network security. This architecture follows real-time industry best practices for scalability, security, and high availability.